Wöden Kusner

Curriculum Vitae, November 2026

wkusner@gmail.com
wkusner.github.io

Research Interests

Applied Geometry and Optimization, Discrepancy and Uniform Distribution, Computer-Assisted Proof, Topology of Configuration Spaces, Topological Data Analysis, Approximation Theory, Representation Theory, Statistical Mechanics and Condensed Matter Physics.

Education

University of Pittsburgh, Pittsburgh, PA USA
Ph.D., Mathematics — August 2014
Advisor: Professor Thomas C. Hales
Dissertation: *Bounds on packing density via slicing*
M.A., Mathematics — December 2010

Haverford College, Haverford, PA USA
B.S., Mathematics — May 2007
Advisor: Professor John J. Flynn
Thesis: *Results in sphere packing density*

Academic Appointments

Director, Quantitative Resource Center — 2025–
Adjunct Assistant Professor of Mathematics
Department of Mathematics
Lawrence University

Instructor of Mathematics — 2023–2025
Department of Mathematics
Lawrence University

Assistant Professor (NTT) — 2020–2022
Department of Mathematics
University of Georgia

Assistant Professor (NTT) — 2017–2020
Department of Mathematics
Data Science Institute Faculty Affiliate
Vanderbilt University

Postdoctoral Associate — 2017–2020
Center for Constructive Approximation
Vanderbilt University

FWF Postdoctoral Researcher — 2014–2017
Institute of Analysis and Number Theory
Graz University of Technology

Visiting Scholar
Isaac Newton Institute (Aberystwyth) — 2023
Fields Institute — 2021
ICERM, Brown University — 2015, 2018
Erwin Schrödinger International Institute, University of Vienna — 2014

Publications

- with R. Kusner. A Gordian pair of links. *Geom. Dedicata*, 2023. arXiv:1908.05610
- with J. Brauchart, P. Grabner, J. Ziefle. Hyperuniform point sets on the sphere: probabilistic aspects. *Monatsh. Math.*, 2020. arXiv:1809.02645
- with J. Brauchart, P. Grabner. Hyperuniform point sets on the sphere: deterministic aspects. *Constr. Approx.*, 2019. arXiv:1709.02613
- with R. Kusner, J. Lagarias, S. Shlosman. Configuration spaces of equal spheres touching a given sphere: the twelve spheres problem. In *New Trends in Intuitive Geometry*, Bolyai Soc. Math. Stud., 2018. arXiv:1611.10297
- with Y. Kallus. The local optimality of the double lattice packing. *Discrete Comput. Geom.*, 2016. arXiv:1509.02241
- On the densest packing of polycylinders in any dimension. *Discrete Comput. Geom.*, 2016. arXiv:1405.0497
- Upper bounds on packing density for circular cylinders with high aspect ratio. *Discrete Comput. Geom.*, 2014. arXiv:1309.6996

Preprints and Work in Progress

- with T. Hales. Packings of regular pentagons in the plane. (*in revision*, *Trans. Amer. Math. Soc.*). arXiv:1602.07220
- with G. Dietler, E. Rawdon, R. Kusner, P. Szymczak. Chirality for crooked curves. (*in revision*). arXiv:2004.10338
- How evenly can four points sit on a sphere? Capture, balance, and the spherical cap discrepancy. (*in preparation*; expository research article intended for the *Amer. Math. Monthly*).
- Spherical cap discrepancy of five to twelve points: optimal configurations, a cross-verified proof, and data. (*in preparation*; intended for *Experimental Math.*).
- Critical radii of kissing configurations: algebraicity, balance, and twelve spheres above $r = 1$. (*in preparation*).
- with G. Buck, R. Kusner. Stopper knots. (*in preparation*).

Software and Data

- `capdisc`: verification code for the optimal spherical cap discrepancy of $N = 5, 6$ points: two independent interval pipelines (Arb and mpmath.iv), a rigorous C mesh, a stand-alone certificate-chain checker, and an atlas of critical configurations for $N = 5$–12, 2026. github.com/wkusner/discrepancy (branch `capdisc`)
- `kisscrit` v0.1.0: certified interval branch-and-bound search and certificate checker for critical kissing configurations (C, FLINT/Arb), 2026. doi:10.5281/zenodo.23603268

Grants and Support

- Isaac Newton Institute visitor support, GeomPack satellite programme (Aberystwyth), 2023
- Workshop funding (co-organizer): Shanks Workshop on Computations and Linear Programming Bounds for Energy, Packing and Covering, Vanderbilt, 2019
- Conference funding (co-organizer): Fundamental Lemma → Discrete Geometry → Formal Verification, Pittsburgh, 2018
- FWF (Austrian Science Fund) postdoctoral position, TU Graz, 2014–2017

Talks

- CTE Lunch and Learn, Lawrence University: *A discussion about the new Quantitative Resource Center and our perspectives of math.* 10/28/25
- Society of Physics Students Journal Club, Lawrence University: *Chirality for crooked curves.* 1/25/24

- AMS Fall Southeastern Sectional, Special Session on Discrete Geometry and Geometric Optimization, South Alabama: *Critical kissing spheres.* 10/15/23
- INI GeomPack satellite programme / 5th International Conference on Packing Problems, Aberystwyth: *Building Gordian links.* 8/22/23
- Carolina Geometry Seminar: *Measuring chirality with the wind.* 4/8/21
- Points Seminar: *Measuring chirality with the wind.* 4/7/21
- Auburn University Colloquium: *Chirality and hydrodynamics.* 11/13/20
- UGA Geometry Seminar: *Building Gordian unlinks.* 11/6/20
- UGA Topology Seminar: *Chirality and hydrodynamics (à la Lord Kelvin).* 8/31/20
- BIRS–CMO Workshop 19w5142, Soft Packings, Nested Clusters, and Condensed Matter, Oaxaca: *Gordian unlinks.* 10/3/19
- Topology and its Applications, WKU: *Gordian configurations (II).* 7/17/18
- ICERM, Brown: *Gordian configurations (I).* 4/11/18
- ICERM, Brown: *Computing discrepancy.* 3/9/18
- Montanuniversität Leoben: *Critical packings & the radius function.* 6/2/17
- CEIM, Universidad de Cantabria: *Critical packings (in the sphere).* 4/22/17
- JMM, Discrete Geometry & Convexity: *Critical packings, rigidity, & the radius function.* 1/6/17
- TU Graz, Fall School: *Critical packings, rigidity, & the radius function.* 9/30/16
- AIM Workshop: *Configurations of spheres.* 9/22/16
- ACG Seminar, Pittsburgh: *Configurations of spheres.* 8/25/16
- MCQMC, Stanford: *Configurations of points w.r.t. discrepancy & uniform distribution.* 8/17/16
- MSRI: *Critical packings, rigidity, & the radius function.* 8/4/16
- AMS Spring Southeastern Sectional, Special Session on New Developments in Discrete and Intuitive Geometry: *Configurations of points w.r.t. discrepancy & uniform distribution.* 3/6/16
- Advanced Topics Seminar, TU Graz: *Configurations of spheres.* 1/22/16
- Zahlentheoretisches Kolloquium, TU Graz: *Problems with packing periodicity.* 12/11/15
- ICERM, Brown: *Can rods pack space more densely than disks pack the plane?* 4/28/15
- ICERM, Brown: *Spherical discrepancy.* 4/9/15
- TU Graz: *Computing spherical cap discrepancy: proof of concept.* 1/22/15
- Guest Lecture, TU Graz: *Introduction to packing problems.* 1/19/15
- Large Structures Seminar, Aalto: *Packing density bounds in higher dimensions.* 11/22/14
- ESI Workshop: *A brief analysis of regular pentagon packings in the plane.* 8/27/14
- Oberwolfach: *Packing polycylinders.* 6/*/14
- Seminar, TU Graz: *Packing density bounds via slicing.* 5/8/14
- Erdős Memorial Lectures, Memphis: *Polycylinder density in higher dimensions.* 3/*/14
- GSS, Pittsburgh: *Some packing problems & an upper bound.* 3/28/13
- A&S Graduate Expo, Pittsburgh: *Packing cylinders with high aspect ratio.* 3/23/13
- ACG Seminar, Pittsburgh: *An upper bound on packing density for circular cylinders with high aspect ratio.* 2/12/13
- Topological Dynamics Workshop, Newton Institute: *Packing circular cylinders.* 11/*/12

Workshops and Research Programs

- Workshop on Institutional Approaches to Quantitative Skill Support, Carleton College. 4/4/25
- ESI Workshop: Optimal Point Configurations on Manifolds (hybrid). 1/22
- 4th International Conference on Packing Problems, Yale. 6/*/19
- Aspen Center for Physics. 6/*/17

– ICERM Workshop, Brown. 9/*/16
– MSRI Summer Research. */16
– Institut Henri Poincaré Workshop. 6/*/16
– IAS–PCMI Researcher: Mathematics and Materials. 6/*/14
– Fields Institute: Workshop in Discrete Geometry. 11/*/13
– IMA Summer School in Topological Methods, Penn. 7/*/11

Teaching

Lawrence University, Appleton, WI USA

Adjunct Assistant Professor 9/25–

- MATH 103: Preparation for Calculus — Fall 26
- MATH 140: Calculus — Spring 26
- MATH 102: Foundations in Math *TA: Ellison McLane* — Winter 26
- MATH 103: Preparation for Calculus — Fall 25
- MATH 591: Random Walks & Electrical Networks — Fall 25

Instructor 9/23–8/25

- MATH 103: Preparation for Calculus — Spring 25
- MATH 103: Preparation for Calculus *TA: Eli Hooker-Reese* — Winter 25
- MATH 591: Differential Topology — Winter 25
- MATH 103: Preparation for Calculus *TA: Elijah Pile* — Fall 24
- MATH 103: Preparation for Calculus — Spring 24
- MATH 102: Foundations in Math *TA: Hasif Ahmed* — Winter 24
- MATH 103: Preparation for Calculus *TA: Mina Mandegar* — Fall 23

Course development: designed the ALEKS-based structure and exercise pack for MATH 103 (2023), later used by colleagues teaching the course.

University of Georgia, Athens, GA USA

Assistant Professor (NTT) 8/20–12/21

- Instructor for MATH 2250: Calculus I (2 sections) — Fall 21
- Instructor for MATH 2260: Calculus II — Spring 21
- Instructor for MATH 2250: Calculus I (2 sections) — Fall 20

Vanderbilt University, Nashville, TN USA

Assistant Professor (NTT) 8/17–8/20

- MATH 2300: Multivariable Calculus — Spring 20
  grader: Emma Francis
- MATH 2300: Multivariable Calculus (2 sections) — Fall 19
  grader: Emma Francis
- Undergraduate Summer Research (3 students) — Summer 19
- MATH 3641/5641: Mathematical Statistics — Spring 19
  grader: Xiaolu Qian
- MATH 2300: Multivariable Calculus — Fall 18
  grader: Xiaolu Qian
- MATH 1010: Probability & Statistical Inference I — Fall 18
  TAs: Jamison Donohue, Min Soo Kim, Shaoyang Zhou; grader: Jessica Wang
- MATH 3641/5641: Mathematical Statistics — Spring 18
- MATH 1011: Probability & Statistical Inference II — Spring 18
  TAs: Junying Fang, Xiayimei Han, Qixuan Wang
- MATH 1010: Probability & Statistical Inference I — Fall 17
  TAs: Amber Teetsel, Qixuan Wang, Helen Zhou

Graz University of Technology, Graz, AT

Lehrbeauftragter 10/14–1/15, 3/16–6/16

Instructor for MAT.670: Packings, Lattices and Configurations Summer 16
Assistant for MAT.902: Höhere Analysis Winter 14

University of Pittsburgh, Pittsburgh, PA USA

Teaching Fellow 9/10–12/11

MATH 0220: Calculus I (2 sections) *E. Arrington, M. Chen* Fall 11
MATH 2700: Graduate Topology *P. Gartside* Fall 11
MATH 1700: Topology *B. Ion* Spring 11
MATH 1410: Foundations of Mathematics *T. C. Hales* Spring 11
MATH 1250: Abstract Algebra II *J. Wheeler* Spring 11
MATH 0230: Calculus II *D. Vassilev* Fall 10
MATH 0220: Calculus I (2 sections) *B. Michael, E. Trofimov* Fall 10

Teaching Assistant 9/09–8/10

Instructor for MATH 0120: Business Calculus Summer 10
MATH 0120: Business Calculus (3 sections) *C. Duane-Lennard, B. Michael* Spring 10
MATH 0240: Calculus III (3 sections) *J. Wheeler, J. Arciero* Fall 09

Mentoring

- Research Mentor (LU Research Fellows), Ellison McLane: discrepancy theory, Summer 2025 (Lawrence)
- Organizer and mentor, Grad School Boot Camp (Math GRE practice exams and application preparation) for Mina Mandegar, Ellison McLane, and Binh Vu, Summer 2025 (Lawrence)
- Directed studies (MATH 591): Differential Topology, Winter 2025; Random Walks and Electrical Networks, Fall 2025 (Lawrence)
- Letters of recommendation for more than 20 students (2018–2025): graduate programs in mathematics, statistics and financial mathematics; summer research and internships (e.g. Caltech SURF, DOE SULI, NREIP); medical school; transfer, study-abroad and off-campus programs; employment
- Research Mentor for Kevin Hu, B.S. Highest Honors 2020 (Vanderbilt)
- Honors Thesis Committee for David K. Zhang, B.S., Founders Medal 2019 (Vanderbilt)
- Dissertation Committee for Oleksandr Vlasiuk, Ph.D. 2018 (Vanderbilt)
- Research Mentor for Jonas Ziefle (Graz)

Certificates

University of Georgia, Athens, GA USA
Certificate in Diversity and Inclusion May 2021

Vanderbilt University, Nashville, TN USA
STEM Teaching Specialization June 2020
Certificate in College Teaching May 2019

Professional Development

- CEL Cohort, Lawrence, 2026–2027
- REFLECT, Lawrence CTE, Winter 2024

Honors and Recognition

- UGA Teaching Academy Fellow, 2021–2022
- Member of the Instructional Corps for the UGA Department of Mathematics, which received the 2022 Regents' Teaching Excellence Award from the University System of Georgia
  University System of Georgia award for departmental teaching excellence.
- Research profiled in *Die Presse* (Science section), 2017
- University of Pittsburgh Honors Convocation, 2013, 2014
- Outstanding Lecture/Presentation, University of Pittsburgh Graduate Expo, 2013
  Award for the best lecture by a graduate student in the Natural Sciences.
- K. Leroy Irvis Fellow, University of Pittsburgh, 2009, 2012
  Competitive university graduate fellowship.
- Bronze Presidential Service Award for AmeriCorps Volunteer Service, 2008

Service

University-focused

- Director: Quantitative Resource Center (Lawrence), 2025–
  Peer tutoring and mentoring of peer tutors, the Quantitative Learning Assistant (QLA) program, quantitative-skills coordination across departments, individualized support for case-managed students, and open office hours for students, faculty, and staff.

Teaching-focused

- Co-organizer: UGA Math Learning and Teaching Seminar, 2022
- UGA Calculus Exam Committee, 2021
- LS-PAC MODELS Mentor, 2021–2023
  National mentoring program for LSAMP-affiliated STEM students.

Research-focused

- Active referee and reviewer for journals and scientific bodies, including BIRS, ESF, NSF, NWO; Ann. of Math., Constr. Approx., DCG, Exp. Math., GCOM, IMRN, JKTR, Proc. Roy. Soc. Edinburgh A, SIAM J. Discrete Math., WADS
- Reviewer: NSF Graduate Research Fellowship Program panel (Mathematical Sciences), 2022
- Co-organizer: AMS Special Session on Discrete Geometry and Geometric Optimization, Fall Southeastern Sectional, 2023 (postponed from 2021)
- Co-organizer: Shanks Workshop (Vanderbilt), 2019
  Proposed, organized, and helped fund the Shanks Workshop on Computations and Linear Programming Bounds for Energy, Packing and Covering.
- Co-organizer: Fundamental Lemma → Discrete Geometry → Formal Verification, 2018
  Proposed, organized, and secured external funding.
- Organizer: Computational Analysis Seminar (Vanderbilt), 2017–2020
  Managed the weekly seminar run out of the Center for Constructive Approximation.
- Website Maintainer: CCA and CA Seminar (Vanderbilt), 2017–2020
  Maintained and updated the website for the Center for Constructive Approximation.
- Organizer: Graduate Seminar in Algebra, Combinatorics and Geometry (Pittsburgh), 2010–2014

Community-focused

- Historic Central Neighborhood Association (Appleton): Secretary, 2021–; Core Team, 2024–
- Edison/Columbus Safe Routes to School Transition Subcommittee (Appleton Area School District), 2025
- Neighborhood Advisory Team, Zion Lutheran Affordable Housing Project (Appleton), 2024
- Fox Valley Data Exchange, data lakehouse specification working group, 2025
- Community Data Hub Project / IFC Data Team (Fox Cities), 2021–2022
  Data team for the Imagine Fox Cities community well-being initiative.
- Kettering Foundation Research Exchange, 2021
- Co-organizer and panelist: BiPOC Narratives in Medicine (Haverford), 2021; Alumni Weekend panelist (Haverford), 2021
- Leadership Committee: Multicultural Alumni Action Group (Haverford), 2020–2024

Graduate student service (Pittsburgh)

- Representative, Dietrich School of Arts and Sciences Council, 2012–2014
- Delegate, Arts and Sciences Graduate Student Organization, 2011–2014
- President, Mathematics Graduate Student Organization, 2013–2014
- Treasurer, Mathematics Graduate Student Organization, 2011–2013
- Treasurer, SIAM University of Pittsburgh Chapter, 2010–2011

Other Works

- *Of Instruments*: working interactive mathematical instruments (slide rules, Napier's bones, planimeters, verniers, and others), 2026
- Untitled work in *Frayed Knot* at The Museum of Everyday Life, 2020
- Cover art for *New Trends in Intuitive Geometry*, 2018

Memberships

AMS, MAA, APS

Technical Skills

C (FLINT/Arb interval arithmetic), Python, Julia, Mathematica; HTML/CSS/JavaScript; Git/GitHub; TEX (LATEX, BIBTEX); AI-assisted research and development (Claude, Claude Code).

Languages

English (native), German (intermediate), Spanish (elementary).

Citizenship

USA

References

Please contact me for references.